Aayush Panda

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Technical Skills

Programming Languages: C, C++, Rust, Python, Java, TypeScript, SQL, Haskell, Scheme, Bash, HTML/CSS, x86 assembly
Machine Learning: Transformers, RNNs, Diffusion models, VAEs, PyTorch, Numpy
Databases: SQL dbs, Firestore, Neo4J, Chroma

Education

University of Waterloo — **GPA:** 3.92/4.00
Bachelor of Computer Science, Honours Co-op — *Sep. 2024 – Apr. 2028*

Experience

ML & NLP Engineering Intern — Sep. 2025 – Present
Huawei Canada
- Developed algorithms for KG generation from natural language documents, with approximate named entity linking
- Developing information-theoretic edit location finding algorithms for LLM-powered text editing
- Adapted a hybrid ASR model architecture to work with natural language context to guide transcription, reduced WER by $\sim 5\%$

Laboratory Assistant — Apr. 2025 – Dec. 2025
University of Waterloo Multi-Sensory Brain and Cognition Lab (MBC)
- Research on how batsmen's eyes track baseballs, measuring visual perception bandwidth, and correlating with batting ability
- Developed hybrid/synthetic data pipelines to train a U-net model to detect baseballs in captured video
- Developed correspondence algorithm to synchronise data from high speed head mounted camera and eye tracking data

Undergraduate Research Assistant — Apr. 2025 – Aug. 2025
University of Waterloo Data Privacy Lab
- Research on private record linkage protocols using locality sensitive hashing, supervised by Dr. Florian Kerschbaum
- Developed a differentially private method for estimating max histogram bin size, for a novel private set intersection algorithm
- Developed k-d cuckoo hashing algorithm to run private set intersection in sub-quadratic time

Software Engineering Intern — Mar. 2025 – Apr. 2025
Toma (YC W25)
- Wrote a service that evalutes dealership customer support quality using an in-house conversational voice AI. This data was then used to automatically shortlist and personalise pitches for ideal clients from over ~ 20k dealerships in the US.
- Worked on model evaluation for Toma's voice agents

Projects

Concord — *Rust, Vulkan, CUDA, HLSL*
- Ongoing project to build a cross-platform runtime in Rust for accelerated computation, with a unified ISA and memory model for CPU/GPU operations.
- Allows for modular bytecode extensions to support custom hardware

Semantify — *Python, React, FastAPI, Sentence Transformers, UMAP, Ollama*
- Built a system that semantically organises **1000+ files** into a directory structure in minutes using embedding models and custom hierarchical clustering and subtree merging algorithms
- Developed an interactive visualization of document embeddings to allow intuitive semantic exploration
- Implemented a RAG-powered chat interface to intelligently query uploaded documents with source citation

Phased Array SONAR — *C++, Python, Xtensa LX6 microprocessor, AVR RISC processors*
- Engineered a sub-degree precision beam-steering phased SONAR array with real-time radar-style display on oscilloscopes.
- Designed a waveguide to reduce inter-element pitch and suppress grating lobes, enhancing beam directivity.
- Built a phased array simulator suite to visualize beamforming, steering, and focus behaviors in 2D.

Patents

CA 3222437: Device for redirection of optical beams using virtual gratings generated by stationary waves.
CA 3119717: Compliant mechanism for operating flight control surfaces of a remotely piloted aircraft.

Awards

University of Waterloo President's Research Award
Jane Street Estimathon @ UWaterloo (2024): First place
Hack the North 2022: Winner (out of 829 participants)
PicoCTF 2022: 2nd place in Canada, 14th (top 0.001%) globally
FIRST Innovation Challenge 2021: Semifinalist